

AdiVaani Hiring Assignment

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Part-1: Classical Neural Machine Translation (Hindi $\longleftrightarrow$ Marathi)

Method 1: BERT-based embeddings

Data pre-processing

- Training Data: 30% of the given training data is used from `train.hi` and `train.mr`. This constituted a training set of the first 72k sentences.
- For validation, 10% of training set *i.e.*, 7k sentences, , are selected from the portion immediately following the first 72k training sentences.

Training Design

- The models used for tokenization and embedding of the data are [Hindi BERT](#) and [Marathi BERT](#).
- The vocabulary size considered is 10000 and the embedding dimension is 256. As the given model consisted of 197285 vocabulary size, a shorter vocabulary size is considered to reduce the size of the embedding matrix with 10000X256, and reduce training time and computation resource usage. The training pipeline of the model is shown in Figure 1.

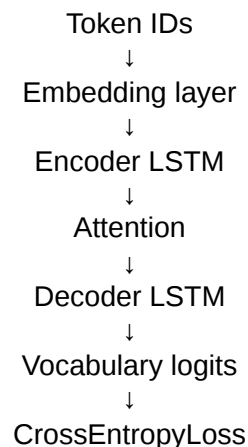


Figure 1: Training pipeline for translation task using BERT-based embeddings.

- For decoding, the beam search decoding strategy is applied. The beam width considered is 5.
- Decoder weight tying is used to share (tie) the weights of the decoder's input embedding matrix with the weights of the decoder's output projection layer, the layer before the softmax linear layer. This is incorporated to reduce the total number of parameters and improve the translation performance.
- The architecture details are
 - Encoder-input-dimension: 768
 - Hidden-dimension: 256
 - Num-Heads: 8
 - Embedding-dim: 256
- Teacher forcing is also used with a ratio of 0.7 for stable training.
- Hyperparameters and Optimization schedule:
 - Epochs: 10
 - Learning Rate Scheduler: 1×10^{-3}
 - Optimizer: Adam
 - Gradient clipping with maximum norm 0.1

Results and Analysis

The training and validation loss for BERT-based embeddings is shown in Figure 2.

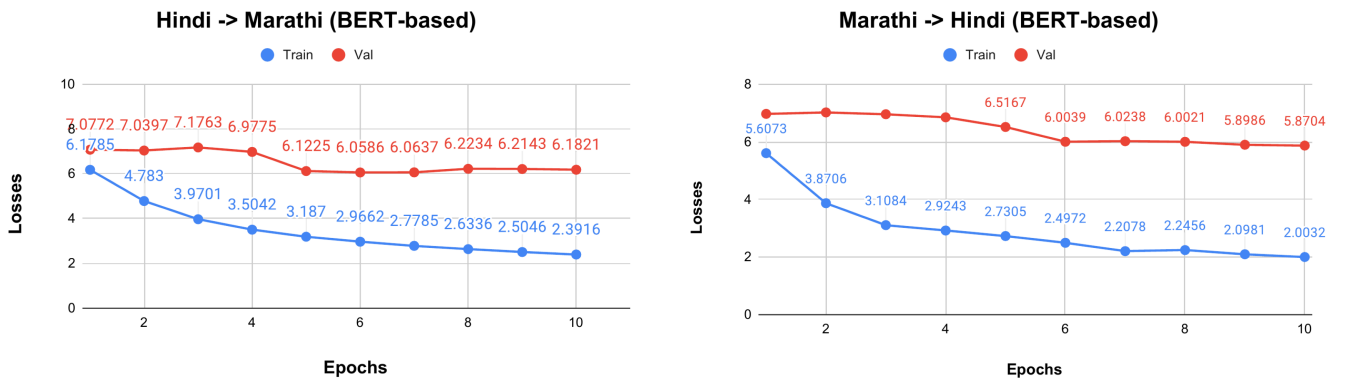


Figure 2: The training and validation loss for BERT-based embeddings.

- The training loss and validation loss decrease over the epochs. The validation loss decreases at a lower rate compared to the training loss.
- It can be observed in the Marathi to Hindi translation that the losses decrease at a higher rate compared to the corresponding losses in Hindi to Marathi translation.

The performance of the models is presented in Figure 3.

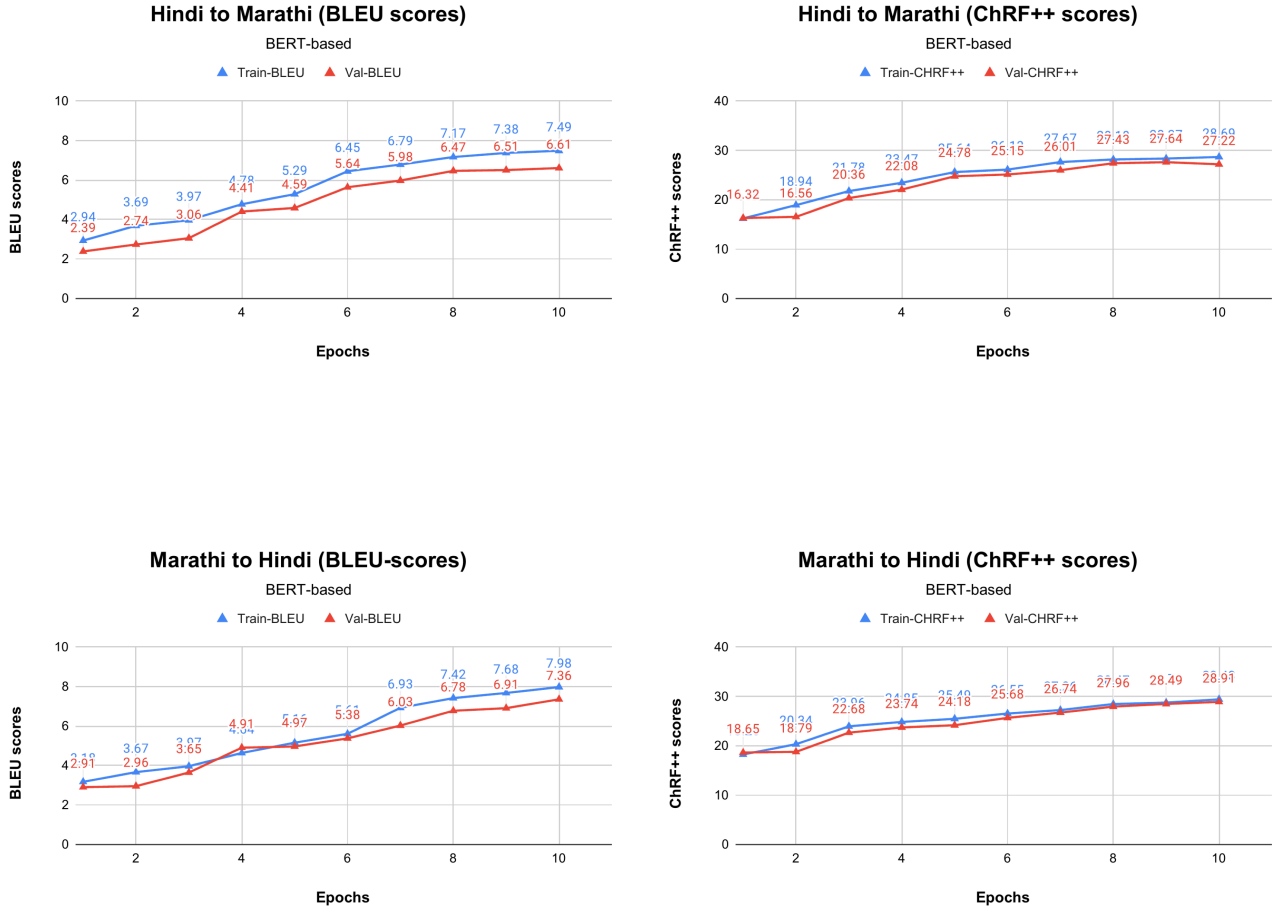


Figure 3: Bleu and ChRF++ scores for the Hindi $\longleftrightarrow$ Marathi translation task with BERT-based embeddings.

- The Bleu scores and ChRF++ scores increase for the translation task in both the training and validation sets.
- Comparatively, Marathi to Hindi yields better performance with respect to both evaluation metrics. The absolute value of the scores in Marathi to Hindi translation is greater compared to the respective scores in Hindi to Marathi translation.
- Marathi is a morphologically richer language (agglutinative) with complex grammatical structures and distinct case markers; translating it to the comparatively less complex Hindi results in better performance.

Method 2: Random Embeddings

The same set-up for training and validation set is chosen as Method §.

- Training Data: 30% of the given training data is used from `train.hi` and `train.mr`. This constituted a training set of the first 72k sentences.
- For validation, 10% of training set *i.e.*, 7k sentences, are selected from the portion immediately following the first 72k training sentences.

Training Design

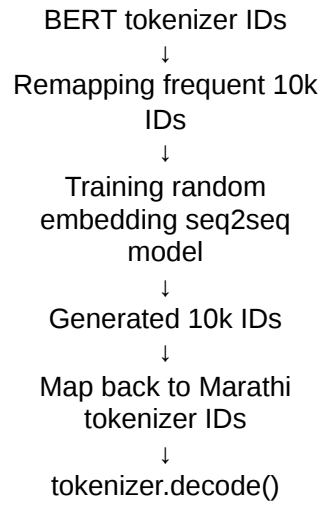


Figure 4: Training pipeline for translation task using randomly initialized embeddings.

- The vocabulary size considered is 10000 and the embedding dimension is 256. As the given model consisted of 197285 vocabulary size, a shorter vocabulary size is considered to reduce the size of the embedding matrix with 10000X256, and reduce training time and computation resource usage. The training pipeline of the model is shown in Figure 4.
- For decoding, the beam search decoding strategy is applied. The beam width considered is 5.
- Decoder weight tying is used to share (tie) the weights of the decoder's input embedding matrix with the weights of the decoder's output projection layer, the layer before the softmax linear layer. This is incorporated to reduce the total number of parameters and improve the translation performance.
- The architecture details are
 - Encoder-input-dimension: 768
 - Hidden-dimension: 256
 - Num-Heads: 8
 - Embedding-dim: 256
- Teacher forcing is also used with a ratio of 0.7 for stable training.
- Hyperparameters and Optimization schedule:
 - Epochs: 10
 - Learning Rate Scheduler: 1×10^{-3}
 - Optimizer: Adam
 - Gradient clipping with maximum norm 0.1
- Compute Resource: 1 GPU - 80 GB NVIDIA A100

Results and Analysis

The training and validation loss for training with random embeddings is shown in Figure 5.

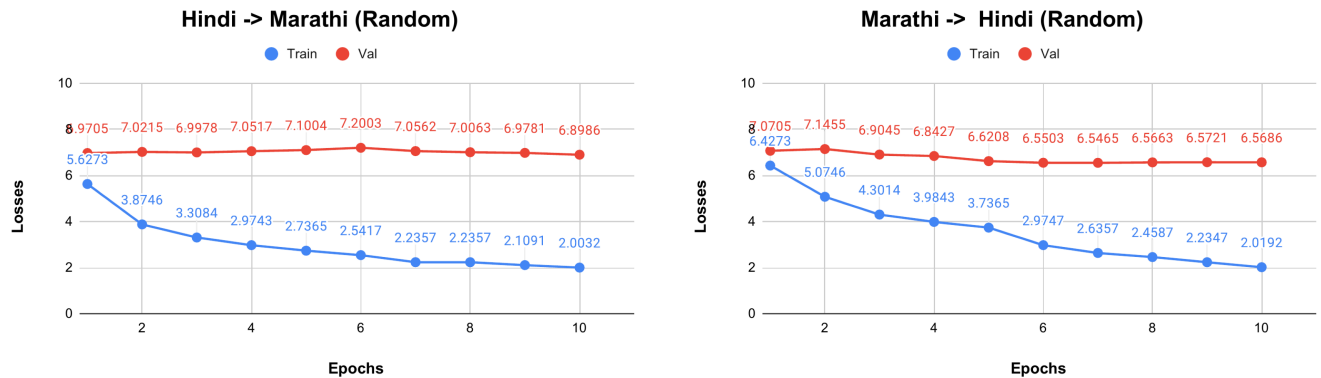


Figure 5: The training and validation loss for training with random embeddings.

- The training loss decreases over the epochs. However, the validation loss remains relatively the same with increasing epochs.
- With close observation, it can be observed that the validation loss in Marathi to Hindi translation decreases slightly with increasing epochs.

The performance of the models is presented in Figure 6.

- The Bleu scores and ChRF++ scores increase for the translation task in both the training and validation sets.
- Comparatively, Marathi to Hindi yields better performance. Marathi is a morphologically richer language (agglutinative) with complex grammatical structures and distinct case markers; translating it to the comparatively less complex Hindi results in better performance.

Comparative Analysis between BERT-based embeddings and Random embeddings

- Comparing the loss plots in Figure 2 and 5, it can be concluded that method 1 with BERT-based embeddings performs better than the randomly initialized embeddings.
- As the BERT embeddings carry the semantic information for the respective languages before the training, the method 1 outperforms the method 2. It can also be observed by comparing Bleu scores and ChRF++ scores in Figure 3 and 6.

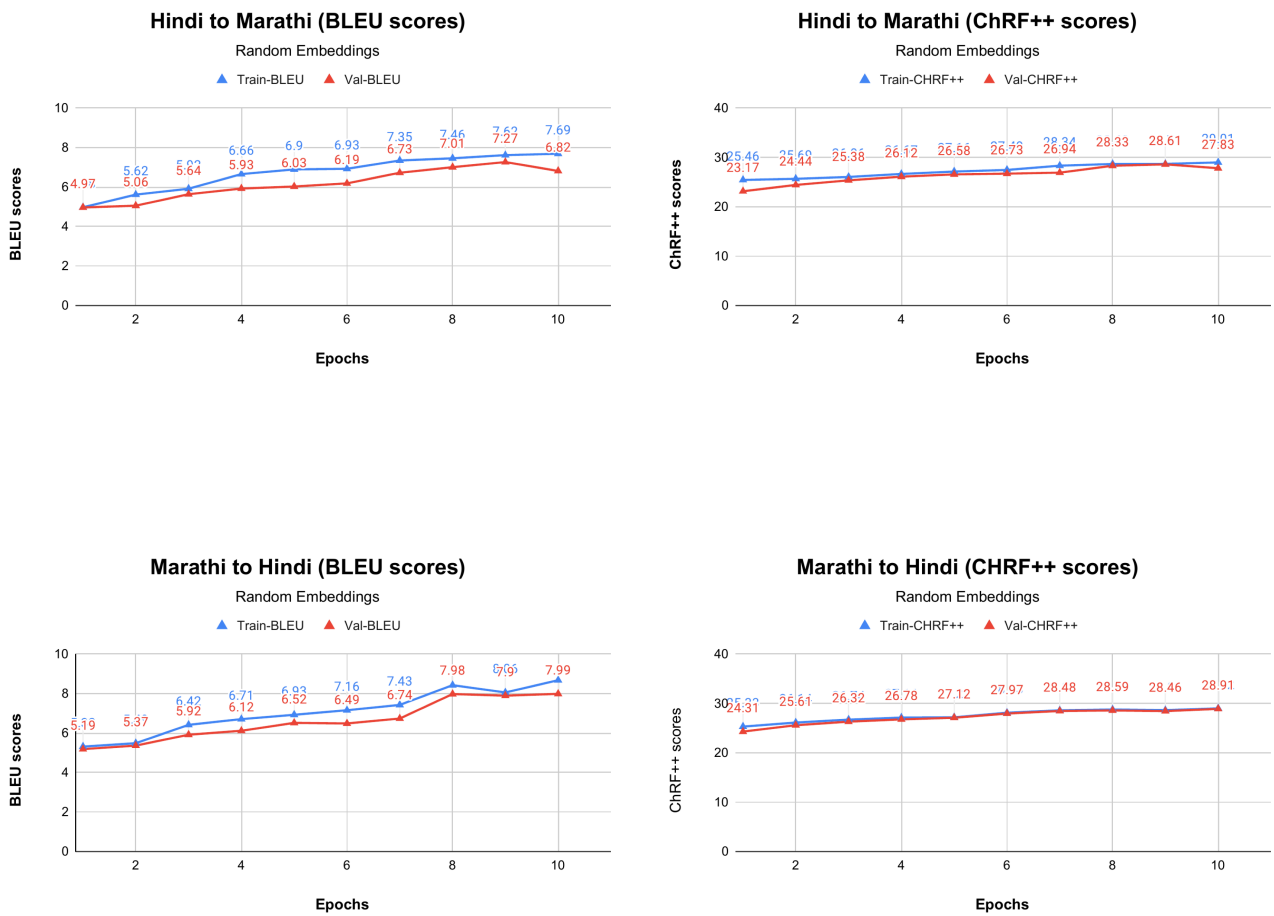


Figure 6: Bleu and ChRF++ scores for the Hindi $\longleftrightarrow$ Marathi translation task with randomly initialized embeddings.